W. Matthew Alampay Davis

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Department of Economics
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Education

Columbia University	University of Oxford	New York University
Dept. of Economics, GSAS	Nuffield College	Stern School of Business
Ph.D Economics (2025)	*M.Phil* Economics (2019)	*B.Sc* Business & Political Economy (2015)

Research

Environmental/Climate Economics · Development Economics · Political Economy

Job market paper

Climate inequality

This paper characterizes the relationship between anthropogenic climate change and global inequality, a subject intersecting two of the defining challenges of the 21st century but which remains virtually unstudied at subnational levels. To make overdue progress addressing this research gap, I improve upon frontier methods for estimating climate impacts to overcome outstanding methodological limitations and apply these refinements to fine-scaled data on globally representative income distributions. Next, I document new evidence that temperature shocks significantly and persistently impact distributions of income within countries, an effect driven by concentrations of harm onto the lowest income-earners in warm climates as well as a surprising vulnerability of the top 1% in these countries to environmental shocks. Integrating these inequality effects over observed distributions of income and the spatial incidence of global warming, I find that climate change between 1981 and 2016 regressively redistributed global shares in a reduced stock of income both between and within countries largely by depriving the world's poorest of economic opportunity that would otherwise have been available. Altogether, these results constitute the most comprehensive evidence yet of the regressive impact of climate change on global inequality.

Work in progress

Temperature, institutions, and the political climate

Despite widespread regard of climate change as a threat to global stability, the nature of the relationship between environmental change, political preferences, and regime change is undertheorized in existing environmental political economy work. I consider two possible frameworks by repurposing influential existing theories of political transition. Under a "mercury uprising" hypothesis, adverse environmental shocks diminish the opportunity cost of contesting autocratic rule thereby improving prospects for democratic reform. On the other hand, a "state of exception" hypothesis suggests that the public responds to these shocks with a willingness to compromise civil liberties, leading to a diminution of democratic institutions.

The merits of these competing theories are empirically tested first by measuring the impacts of identified temperature shocks on survey-based proxies for the demand for democracy. Next, I use multi-valued and dichotomous measures of democracy to investigate the dynamic effects of these shocks on the democratic quality of institutions. The results from these analyses consistently support the latter theory. A particularly striking finding is that a positive 1°C temperature shock increases the propensity for autocracies in temperate geographies to democratize by between 0.030-0.075 percentage points depending on the operating definition of democracy but reduces this propensity for their tropical counterparts by 0.032-0.048 percentage points. The same shocks increase the propensity for warm-climate democracies to break down by 0.011-0.040 percentage points. Finally, we find that the same shocks have null effects on the propensity for temperate democracies to undergo reversal, suggesting these institutions have historically proven impervious to environmental disruption. These findings highlight the urgency of climate change as a means of safeguarding global democracy, particularly in the Global South.

Evidence of a drought effect on the hazard into spousal violence

with Tanushree Goyal

In this ongoing work, we explore how income shocks may impact the probability that a previously non-violent marriage turns violent. We combine high-resolution weather data with comprehensive duration data on the experiences with domestic violence of married women in India to characterize how drought-induced shocks may affect gendered outcomes. Early findings suggest that negative income shocks substantially reduce the hazard into marital violence. Specifically, a negative precipitation shock of a magnitude expected once every 4-5 years reduces the risk of a never-violent relationship becoming violent by 0.037 percentage points, representing a 70% reduction in the baseline hazard during the formative years of a marriage. Secondary results find that the practice of child marriage may be less economically motivated than marriages occurring later in life, suggesting economic growth is unlikely to mitigate these persistent patriarchal practices.

Elite capture and identity politics: A formal model of substantive representation

Publications

Large potential reduction in economic damages under UN mitigation targets
with Marshall Burke, Noah S. Diffenbaugh *Nature* (2018)

We present a probabilistic framework for assessing aggregate economic impacts of anthropogenic warming. Our construction decomposes uncertainty associated with mid-century and end-of-century country-level projections into error associated with i) econometric estimation of the economic effects of environmental change, ii) climate models of the spatial distribution of anthropogenic warming, iii) the projected schedule of greenhouse gas concentrations associated with radiative forcings, and iv) the social discounting regime of choice. We apply this framework to characterize the economic benefits of climate policy, emphasizing how achieving the most ambitious mitigation targets of the 2015 Paris Agreement would obviate essentially certain economic calamity concentrated in developing countries.

Combining satellite imagery and machine learning to predict poverty
with Neal Jean, Marshall Burke, Michael Xie, David B. Lobell, Stefano Ermon *Science* (2016)

We demonstrate a scalable method to address the scarcity and prohibitive expense of collecting disaggregated welfare data in the developing world by extracting economic information from an unconventional but abundant data source with increasingly frequent and near-global coverage: daytime satellite photography. Our "transfer learning" pipeline proceeds by first assigning a convolutional neural network model pre-trained for generic image classification the task of identifying features in georeferenced satellite images predictive of night-time luminosity, a crude proxy for economic activity. In effect, the CNN learns to produce a nonlinear mapping from high-resolution unstructured image data to low-dimensional vector representations of their most economically relevant features. Ridge regression models are then trained on survey data to refine the feature selection for out-of-sample predictive performance. In an exploratory application to five diverse sub-Saharan African countries—Nigeria, Tanzania, Uganda, Malawi, and Rwanda—our open-source models trained on publicly available data are found to explain up to 75% of the variation in village-level consumption and asset wealth measured by traditional household surveys.

Referee service *Journal of Development Economics (2)*

Teaching

Graduate Teaching Fellow at Columbia University (nine semesters, 2020–2024)
Research consultant to all UG/MA economics thesis writers and research assistants
ECON-4999 – Senior honors thesis workshops (design and instruction)
ECON-3412 – Introduction to econometrics (TA)
ECON-3211 – Intermediate microeconomics (TA)
ECON-2257 – The global economy (TA)

Past affiliations

Research associate at Climate Econometrics (2017–2019)
Nuffield College, University of Oxford
supervised by David F. Hendry, Elizabeth Baldwin

Pre-doctoral scholar at Center on Food Security and the Environment (2015–2017)
Dept. of Earth System Science, Stanford University

supervised by Marshall Burke, David Lobell

Honors

Program for Economic Research Summer Fellowship (2024)
Columbia economics department award supporting summer research activities

Wueller Fourth-Year Pre-Dissertation Award (2023)
Columbia economics department runner-up best dissertation proposal

UC Berkeley and Sloan Foundation Diversity Fellowship (2022)
Awarded to five doctoral students in environmental and energy economics

Columbia GSAS Dean's Fellowship and Dissertation Fellowship (2019)
Full-tuition scholarship and living stipend for up to six years of graduate study at Columbia

Oxford M.Phil top-scoring economics thesis (2019)
Department's first 'exceptional' (mark of 80+) thesis in at least four years

Nuffield College–Department of Economics Joint Studentship (2017)
Full-tuition scholarship and living stipend for up to four years of graduate study at Oxford

Placement

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Personal

Background	Philippines (nationality) · Indonesia (13 years) · Vietnam
Study and career	United States · United Kingdom · China
Immediate family	Philippines · Australia · Canada